

Emilio Luis Sáenz Guillén

Ph.D. Candidate · Bayes Business School

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EDUCATION /

CERTIFICATIONS

BAYES BUSINESS SCHOOL **M.PHIL./PH.D. IN ACTUARIAL** **SCIENCE**

2022 - Present

1ST YEAR COURSES

Bayes Business School

Probability and Mathematical Statistics (SMM941); Financial Econometrics (SMM943); Pensions and Other Benefits (SMM033; SP4); Machine Learning (SMM636); Applied Deep Learning (SMM768).

London School of Economics and Political Science

Probability and Mathematical Statistics I (ST552); Probability and Mathematical Statistics II (ST553).

ADVANCE HIGHER **EDUCATION (UK)**

ASSOCIATE FELLOW (AFHEA)

May 2024.

CONSEJO PROFESIONAL **DE CIENCIAS ECONÓMICAS** **DE LA CIUDAD AUTÓNOMA** **DE BUENOS AIRES**

FULLY QUALIFIED ACTUARY

Dec 2023.

UNIVERSITY OF NAVARRA

MASTER IN ECONOMICS AND **FINANCE (MEF)**

2020 - 2021. GPA: 9.28/10

Master's Thesis:

The Chilean Pension System: Actuarial Analysis of a Paradigmatic Social Security Program.

Supervisors: Adj. Prof. José Azar (Ph.D. in Economics, Princeton University) and Dr. Act. Eduardo Melinsky (Director of the Actuarial Program & Research Centre for Insurance, University of Buenos Aires).

SUMMARY

I am an Actuary and Economist, with a deep-rooted interest in the development of statistical methods with applications in the context of finance and actuarial science. I am pursuing a Ph.D. in Actuarial Science at Bayes Business School. My research to date has focused on stochastic modelling using spline functions and copulas, including work on generalised additive models, gradient boosting, and non-parametric density estimation, as well as on modelling real and perceived income, standard of living, and their implications in marketing. Prior to my Ph.D., I have worked as Research Assistant examining the impact of stakeholders' ESG pressures on banking behaviour, and the potential moral hazards introduced by government bailout guarantees. In a previous engagement, I worked jointly with the Central Bank of the Argentine Republic (BCRA), studying the effect of network centrality on interest rate spreads in Argentine inter-bank markets.

ACADEMIC EXPERIENCE

RESEARCH

BAYES BUSINESS SCHOOL | DOCTORAL RESEARCHER

2022 - Present | London, United Kingdom

- Research topics: splines; gradient boosting methods; micro-level projection of mortality; pension systems modelling.
- Supervisors: Prof. Vladimir Kaishev (Ph.D. in Statistics and Information Theory, Moscow Technical University) and Dr. Douglas Wright (Ph.D. in AM&S, Heriot-Watt University).

IESE BUSINESS SCHOOL | RESEARCH ASSISTANT

Jul 2021 – Jul 2022 | Barcelona, Spain

- Worked under the supervision of Assoc. Prof. Christian Eufinger (Ph.D. in Finance, Goethe University), Assoc. Prof. Stefano Sacchetto (Ph.D. in Economics, London Business School) and Assist. Prof. Valentina Raponi (Ph.D. in Finance, Imperial College Business School and Ph.D. in Methodological Statistics, University of Rome La Sapienza).
- Research topics: banking and corporate finance.

INSTITUTO INTERDISCIPLINARIO DE ECONOMÍA POLÍTICA DE **BUENOS AIRES (IIEP-BAIRES) | RESEARCH ASSISTANT**

2019 - 2020 | Buenos Aires, Argentina

- *Econometric methods for the selection of levels of data grouping.* - Prof. Gabriel Montes-Rojas, Ph.D. in Economics, University of Illinois Urbana-Champaign.

INSTITUTO DE INVESTIGACIÓN SOCIAL, ECONÓMICA Y **POLÍTICA CIUDADANA (ISEPCI) | RESEARCHER**

2018 | Buenos Aires, Argentina

- Índice Barrial de Precios (IBP).

UNIVERSITY OF BUENOS AIRES

ACTUARY DEGREE

Society of Actuaries (SOA)
Recognition Tier: UCAP-AC;
Level of Actuarial Courses
Offered: Masters
2018 - 2020.

BACHELOR'S DEGREE IN ECONOMICS

2013 - 2018. Cum Laude.

COMPUTING

Maintainer – GeDS R Package
CRAN; GitHub

Other programming & software

- Python • MATLAB • Stata • LaTeX
- SQL • Bejerman Accounting System
- Office (Word, Power Point, Excel)

LANGUAGES

English

Fluent. 106/120 in TOEFL; First
Certificate in English.

French

Fluent. Dalf C1 Alliance Française.

German

Beginner. Fit in Deutsch 2,
Goethe Institut.

Portuguese

Bilingual, 7 years living in Portugal.

Basque

Fluent.

Catalan

Mother tongue.

Spanish

Mother tongue.

OTHER CERTIFICATES

Introductory Certificate in Academic Practice

Learning Enhancement and
Development (LEaD). City, University
of London.

GRE

Quantitative reasoning: 166/170,
86th percentile.

TEACHING

BAYES BUSINESS SCHOOL | TEACHING ASSISTANT

Sep 2023 - Present | London, United Kingdom

- Probabilistic Modelling (AS3209) - Prof. Vladimir Kaishev (Ph.D. in Statistics and Information Theory, Moscow Technical University).
- Mathematics for Actuarial Science (AS1056) - Dr. Russell Gerrard (Ph.D. in Stochastic Processes, Cambridge University).

IMPERIAL COLLEGE BUSINESS SCHOOL | POSTGRADUATE

TEACHING ASSISTANT

Sep 2024 - Present | London, United Kingdom

- Finance and Pricing Strategy - MSc Strategic Marketing, Prof. Mushegh Harutyunyan (Ph.D. in Economics, Washington University in St. Louis).

LONDON BUSINESS SCHOOL | POSTGRADUATE TEACHING ASSISTANT

Feb - April 2023/24 | London, United Kingdom

- Time Series Analysis (E334; MBA programme) - Prof. Derek Bunn (Ph.D. in Economics, London Business School).

UNIVERSITY OF NAVARRA | TEACHING ASSISTANT

September 2020 - June 2021 | Pamplona, Navarra, Spain

- Marketing II and Family Business - Assoc. Prof. Goretti Cabaleiro, Ph.D. in Business Economics and Quantitative Methods, University Carlos III.
- Strategic Management - Assoc. Prof. Ricardo Mateo Dueñas, Ph.D. in Business Administration and Accounting, University of Navarra.

UNIVERSITY OF BUENOS AIRES | TEACHING ASSISTANT

2020 | Buenos Aires, Argentina

- Econometrics I - Prof. Gabriel Montes-Rojas, Ph.D. in Economics, University of Illinois Urbana-Champaign.

CONFERENCES

2025

- 28th International Congress on Insurance: Mathematics and Economics held in Tartu, Estonia, on July 1-4, 2025. *Nonparametric Free-Knot Spline Density and Distribution Estimation for Heavy-Tailed Data*
- Insurance Data Science Conference, 19-20 June 2025, Bayes Business School, City St George's, University of London. *Non-Parametric Insurance Loss Modelling using Variable-Knot Splines*.
- Perspectives on Actuarial Risks in Talks of Young Researchers 2025 (Liverpool, UK). *Estimation of Probability Density and Distribution Functions using Variable-Knot Splines with Applications to Insurance Loss Data*.

2024

- Bayes Business School - KU Leuven - University of Amsterdam PhD Workshop 2024. *Simultaneous Estimation of Probability Density and Distribution Functions using Variable-Knot Splines*
- RSS International Conference 2024, 2-5 September, Brighton (UK). *Augmented Spline Regression for Advanced Data Analysis: Generalized Additive Models and Functional Gradient Boosting with Geometrically Designed (GeD) Splines*.
- 26th International Conference on Computational Statistics (COMPSTAT 2024), 27-30 August 2024, University of Giessen. *Enhancing Geometrically Designed Spline (GeDS) regression through Generalized Additive Models and Functional Gradient Boosting*.
- Insurance Data Science Conference, 17-18 June 2024, Stockholm University, Department of Mathematics. *Generalized Additive Models & Functional Gradient Boosting with Geometrically Designed (GeD) Splines: Application to Insurance Data*.

VOLUNTEERING

2013 - 2017 Solidarity work in Villa La Cava, San Isidro, Buenos Aires.

2011 Solidarity work in ONG Juan Bonal España.

OTHER WORK EXPERIENCE

GRUPO ISN ARGENTINA, PURCHASING AND FINANCE

DEPARTMENT | MANAGER

May 2017 - Jul 2018 | Buenos Aires, Argentina

Manager of the Department of Purchasing and Finance during the launch of a new contract with Volkswagen.

Collections | Purchases (market survey, budget evaluation, dealing with suppliers) | General accounting.

UNIVERSITY OF BUENOS AIRES | ADMINISTRATOR

Oct 2015 - Dec 2016 | Buenos Aires, Argentina

Administrative assistant at the Student Center of the Faculty of Psychology of the University of Buenos Aires.

Responsibilities included dealing with suppliers, payment of wages, allocation of scholarships and organization of academic seminars. This was a part-time position.

EL MOLINO RESTO/BAR | Cook

Feb 2014 - Jun 2015 | Buenos Aires, Argentina

Cook at grill restaurant. Part-time job.

SAUMA WAGEN ARGENTINA | INTERNSHIP

2012 | Buenos Aires, Argentina

100-hour accounting internship at the official Ford and Volkswagen dealership.

SUPERVISION IN ARGENTINIAN ELECTIONS PASO and Final 2013, 2015, 2017 and 2019.

AWARDS, FELLOWSHIPS, & GRANTS

2019 | UBACyT research scholarship, University of Buenos Aires